

Supplementary Appendix to “Testing for Equal Average Forecast Accuracy in Possibly Unstable Environments”

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August 21, 2024

Appendix A: Proof of theorems

In Appendix A and B, $\mathbb{C}$ denotes a generic large positive number. We also denote $\varepsilon_t \equiv d_t - m_t$, and from our assumptions it is clear that ε_t will be a zero mean sequence satisfying the dependence and moment conditions in Assumption 2.

Proof of Theorem 1

The estimator can be decomposed as follows

$$\begin{aligned}
 \hat{\Omega} &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n (d_t - \bar{d})(d_s - \bar{d}) k\left(\frac{t-s}{b}\right) \\
 &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n d_t d_s k\left(\frac{t-s}{b}\right) - 2n^{-1} \bar{d} \sum_{t=1}^n \sum_{s=1}^n d_t k\left(\frac{t-s}{b}\right) \\
 &\quad + n^{-1} (\bar{d})^2 \sum_{t=1}^n \sum_{s=1}^n k\left(\frac{t-s}{b}\right) \\
 &\equiv A_1 + A_2 + A_3.
 \end{aligned}$$

The term A_2 can be analysed as follows

$$\begin{aligned}
 A_2 &= -2n^{-1} \bar{d} \left(\sum_{l=-(n-1)}^{(n-1)} k\left(\frac{l}{b}\right) \sum_{t=1 \vee (1+l)}^{n \wedge (n+l)} d_t \right) \\
 &= -2(\bar{d})^2 \sum_{l=-(n-1)}^{(n-1)} k\left(\frac{l}{b}\right) - 2n^{-1} \bar{d} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \left(\sum_{t=n-l+1}^n d_t + \sum_{t=1}^l d_t \right) \\
 &\equiv A_{21} + A_{22}.
 \end{aligned}$$

Using Lemma 1 and Lemma 2, $A_{21} = -2(k(0) + 2b\lambda_k) \left(\int_0^1 m(x)dx \right)^2 + o_p(1)$. Noticing that $E|d_t| < \infty$ for all t , A_{22} satisfies

$$E|A_{22}| \leq \mathbb{C}n^{-1}\bar{d} \sum_{l=1}^{(n-1)} \left| k\left(\frac{l}{b}\right) \right| l = O(b^2/n) = o(1)$$

which can be derived in a similar way to Lemma 1. In total,

$$A_2 = -2(2b\lambda_k + k(0)) \left(\int_0^1 m(x)dx \right)^2 + o_p(1).$$

The term A_3 can be analysed in the same way as A_{21} to obtain

$$A_3 = (k(0) + 2b\lambda_k) \left(\int_0^1 m(x)dx \right)^2 + o_p(1).$$

Now, A_1 can be further decomposed as

$$\begin{aligned} A_1 &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n (m_t + \varepsilon_t)(m_s + \varepsilon_s) k\left(\frac{t-s}{b}\right) \\ &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n m_t m_s k\left(\frac{t-s}{b}\right) + 2n^{-1} \sum_{t=1}^n \sum_{s=1}^n m_t \varepsilon_s k\left(\frac{t-s}{b}\right) \\ &\quad + n^{-1} \sum_{t=1}^n \sum_{s=1}^n \varepsilon_t \varepsilon_s k\left(\frac{t-s}{b}\right) \\ &\equiv A_{11} + A_{12} + A_{13}. \end{aligned}$$

Next, A_{11} can be rewritten as follows

$$\begin{aligned} A_{11} &= n^{-1}k(0) \sum_{t=1}^n m_t^2 + 2n^{-1} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \sum_{t=1}^{n-l} m_t m_{t+l} \\ &= \left(k(0) + 2 \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \right) n^{-1} \sum_{t=1}^n m_t^2 + 2n^{-1} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \left(\sum_{t=1}^{n-l} m_t m_{t+l} - \sum_{t=1}^n m_t^2 \right). \end{aligned}$$

Using Lemma 1 and similar proof as above, we find that the first term above satisfies

$$\left(k(0) + 2 \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \right) n^{-1} \sum_{t=1}^n m_t^2 = (k(0) + 2b\lambda_k) \left(\int_0^1 m^2(x)dx \right) + o_p(1).$$

By Lemma 3 the second term of the decomposition of A_{11} satisfies

$$\left| 2n^{-1} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \left(\sum_{t=1}^{n-l} m_t m_{t+l} - \sum_{t=1}^n m_t^2 \right) \right| \leq 2\mathbb{C}n^{-1} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) l = O\left(\frac{b^2}{n}\right) = o(1).$$

Therefore,

$$A_{11} = (k(0) + 2b\lambda_k) \left(\int_0^1 m^2(x) dx \right) + o_p(1).$$

The term A_{12} can be rewritten as

$$A_{12} = 2n^{-1}k(0) \sum_{t=1}^n m_t \varepsilon_t + 2n^{-1} \sum_{l=1}^{n-1} k\left(\frac{l}{b}\right) \sum_{t=1}^{n-l} (m_t \varepsilon_{t+l} + \varepsilon_t m_{t+l}).$$

First notice that $n^{-1} \sum_{t=1}^{n-l} m_t \varepsilon_{t+l}$ is a sum of an α -mixing sequence and its moment can be evaluated using Proposition 2.8 of Fan and Yao (2008) to obtain $E \left| n^{-1} \sum_{t=1}^{n-l} m_t \varepsilon_{t+l} \right| = O(n^{-1/2})$ for all $l \geq 0$. In the same way $E \left| n^{-1} \sum_{t=1}^{n-l} \varepsilon_t m_{t+l} \right| = O(n^{-1/2})$. Together with Lemma 1 we have $A_{12} = O_p(b/n^{1/2}) = o_p(1)$. For A_{13} , Theorem 2.1 of De Jong and Davidson (2000) implies $A_{13} \xrightarrow{p} \Omega$.

To summarize, $(A_{11} + A_2 + A_3) = (k(0) + 2b\lambda_k) \left(\int_0^1 m^2(x) dx - \left(\int_0^1 m(x) dx \right)^2 \right) + o_p(1)$, $A_{12} = o_p(1)$ and $A_{13} \xrightarrow{p} \Omega$, and the result of the theorem follows.

Proof of Theorem 2

For the DM statistic

$$DM = \sqrt{n} \frac{1}{n} \sum_{t=1}^n \frac{(m_t + \varepsilon_t)}{\sqrt{\hat{\Omega}}} = \sqrt{n} \frac{\frac{1}{n} \sum_{t=1}^n m_t}{\sqrt{\hat{\Omega}}} + n^{-1/2} \frac{\sum_{t=1}^n \varepsilon_t}{\sqrt{\hat{\Omega}}} \equiv D_1 + D_2.$$

(i) Under H_0 and when $m(x)$ is constant, in view of Theorem 1, Lemma 2 and Theorem 5.20 of White (2001), we have $D_1 = o_p(1)$ and $D_2 \xrightarrow{d} N(0, 1)$, and it follows that $DM \xrightarrow{d} N(0, 1)$ so that $P(|DM| > z_{1-\alpha/2}) \rightarrow \alpha$. When $m(x)$ is time varying, by Theorem 1, $\hat{\Omega} = O_p(b)$, so both D_1 and D_2 are $o_p(1)$ and it follows that $DM \xrightarrow{p} 0$ and $P(|DM| > z_{1-\alpha/2}) \rightarrow 0$.

(ii) Under H_1 , when $m(x)$ is constant, it is straightforward to see that $D_1 = O_p(\sqrt{n})$ and $D_2 \xrightarrow{d} N(0, 1)$, and it follows that $DM = O_p(\sqrt{n})$ and $P(|DM| > z_{1-\alpha/2}) \rightarrow 1$. When $m(x)$ is time varying, $\hat{\Omega} = O_p(b)$ so $D_1 = O_p(\sqrt{n/b})$ and $D_2 = o_p(1)$, and it follows that $DM = O_p(\sqrt{n/b})$ and $P(|DM| > z_{1-\alpha/2}) \rightarrow 1$ because $n/b \rightarrow \infty$.

Proof of Theorem 3

Under the local alternatives H_{1n} , it can be shown that the result of Theorem 1 is still formally valid using the proof strategy thereof. Therefore, when $m(x)$ is constant and $\gamma_n = n^{-1/2}$,

$$D_1 = \sqrt{n} \frac{\frac{1}{n} \sum_{t=1}^n (m_t^0 + \gamma_n m_t^1)}{\sqrt{\hat{\Omega}}} \xrightarrow{p} \frac{\int_0^1 m^1(x) dx}{\sqrt{\Omega}} \equiv \Delta'$$

and $D_2 \xrightarrow{d} N(0, 1)$ still. It follows that $DM \xrightarrow{d} N(\Delta', 1)$ and the claimed local power function follows.

When $m(x)$ is time varying and $\gamma_n = (n/b)^{-1/2}$, we have

$$D_1 \xrightarrow{p} \frac{\int_0^1 m^1(x) dx}{\sqrt{(2 \int_0^\infty k(u) du) \left(\int_0^1 (m^1(x))^2 dx \right)}} \equiv \Delta$$

and $D_2 = o_p(1)$. It then follows that $DM \xrightarrow{p} \Delta$, which is deterministic. Therefore the local power function of the DM test takes either the value 0 or 1, depending on the relative magnitude between the chosen critical value $z_{1-\alpha/2}$ and the limit Δ , as claimed by the theorem.

Proof of Theorem 4

First note that

$$\begin{aligned} \hat{\Omega}' &= \sum_{t=1}^n \sum_{s=1}^n (m_t - \hat{m}_t + \varepsilon_t)(m_s - \hat{m}_s + \varepsilon_s) k\left(\frac{t-s}{b}\right) \\ &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n (m_t - \hat{m}_t)(m_s - \hat{m}_s) k\left(\frac{t-s}{b}\right) \\ &\quad + n^{-1} \sum_{t=1}^n \sum_{s=1}^n (m_t - \hat{m}_t) \varepsilon_s k\left(\frac{t-s}{b}\right) + n^{-1} \sum_{t=1}^n \sum_{s=1}^n \varepsilon_t \varepsilon_s k\left(\frac{t-s}{b}\right) \\ &\equiv D_1 + D_2 + D_3. \end{aligned}$$

Let $\bar{m}_t \equiv \sum_{s=1}^n w_{t,s} m_s$. Then, D_1 can be analysed as follows

$$\begin{aligned}
|D_1| &\leq n^{-1} k(0) \sum_{t=1}^n (m_t - \hat{m}_t)^2 + 2n^{-1} \sum_{l=1}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \sum_{t=1}^{n-l} |(m_t - \hat{m}_t)(m_{t+l} - \hat{m}_{t+l})| \\
&\leq n^{-1} k(0) \sum_{t=1}^n (m_t - \hat{m}_t)^2 + n^{-1} \sum_{l=1}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \left(\sum_{t=1}^{n-l} (m_t - \hat{m}_t)^2 + \sum_{t=1}^{n-l} (m_{t+l} - \hat{m}_{t+l})^2 \right) \\
&\leq n^{-1} k(0) \sum_{t=1}^n (m_t - \hat{m}_t)^2 + 2n^{-1} \sum_{l=1}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \left(\sum_{t=1}^n (m_t - \hat{m}_t)^2 \right) \\
&\leq \mathbb{C} n^{-1} \left(\sum_{l=0}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \right) \left(\sum_{t=1}^n (m_t - \hat{m}_t)^2 \right) \\
&\leq 2\mathbb{C} n^{-1} \left(\sum_{l=0}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \right) \sum_{t=1}^n (\bar{m}_t - \hat{m}_t)^2 + 2\mathbb{C} n^{-1} \left(\sum_{l=0}^{n-1} \left| k\left(\frac{l}{b}\right) \right| \right) \sum_{t=1}^n (\bar{m}_t - m_t)^2.
\end{aligned}$$

Using Lemma 1 and Lemma 4, the first term is $O_p(n^{-1}bh^{-1}) = O_p\left(\frac{b}{nh}\right)$ which is $o_p(1)$ under Assumption 6. Using Lemma 6, the second term has order $O(n^{-1}bnh) = O(bh) = o(1)$ by Assumption 6. Therefore $D_1 = o_p(1)$.

Next, D_2 can be decomposed as

$$\begin{aligned}
D_2 &= n^{-1} \sum_{t=1}^n \sum_{s=1}^n (m_t - \bar{m}_t) \varepsilon_s k\left(\frac{t-s}{b}\right) + n^{-1} \sum_{t=1}^n \sum_{s=1}^n (\bar{m}_t - \hat{m}_t) \varepsilon_s k\left(\frac{t-s}{b}\right) \\
&\equiv D_{21} + D_{22}.
\end{aligned}$$

First consider D_{21} . Note that since $m_t - \bar{m}_t$ is deterministic, D_{21} can be analysed in the same way as the term A_{12} in the proof of Theorem 1 to obtain $D_{21} = o_p(1)$. The D_{22} term can be written as

$$D_{22} = n^{-1} \sum_{t=1}^n \sum_{s=1}^n \sum_{j=1}^n w_{t,j} \varepsilon_j \varepsilon_s k\left(\frac{t-s}{b}\right).$$

We then have

$$\begin{aligned}
E|D_{22}| &\leq n^{-1} \sum_{t=1}^n \sum_{s=1}^n \sum_{j=1}^n E|\varepsilon_j \varepsilon_s| w_{t,j} \left| k\left(\frac{t-s}{b}\right) \right| \\
&\leq n^{-1} \max_{1 \leq t, j \leq n} w_{t,j} \sum_{s=1}^n \sum_{j=1}^n E|\varepsilon_j \varepsilon_s| \sum_{t=1}^n \left| k\left(\frac{t-s}{b}\right) \right| \\
&\leq \mathbb{C} n^{-1} \max_{1 \leq t, j \leq n} w_{t,j} \sum_{s=1}^n \sum_{j=1}^n |j-s|^{-1-\varepsilon \times (1-2/r)} \sum_{l=-(n-1)}^{n-1} \left| k\left(\frac{l}{b}\right) \right|
\end{aligned}$$

where we have used the covariance inequality for an α -mixing sequence and the non-negativity of $|k(\frac{l}{b})|$. Now, by noting that $\max_{1 \leq t, j \leq n} w_{t,j} = O(1/(nh))$, $\sum_{s=1}^n \sum_{j=1}^n |j-s|^{-1-\varepsilon \times (1-2/r)} = O(n^{\varepsilon \times (1-2/r)})$, we have

$s|^{-1-\varepsilon \times (1-2/r)} = O(n)$ and $\sum_{l=-(n-1)}^{n-1} \left| k\left(\frac{l}{b}\right) \right| = O(b)$, we have $E|D_{22}| = O(b/(nh)) = o(1)$ and it follows that $D_{22} = o_p(1)$. Therefore $D_2 = o_p(1)$.

As in the proof of Theorem 1, $D_3 \xrightarrow{p} \Omega$, and the proof is then complete.

Proof of Theorem 5

The DM' statistic can be decomposed as

$$DM' = \sqrt{n} \frac{1}{n} \sum_{t=1}^n \frac{(m_t + \varepsilon_t)}{\sqrt{\hat{\Omega}'}} \equiv D'_1 + D'_2.$$

Since $\hat{\Omega}' \xrightarrow{p} \Omega$ regardless of the time varying nature of the $m(x)$ function by Theorem 4, the claimed result of the theorem follows using the same argument used in the proof of Theorem 2.

Proof of Theorem 6

Under the local alternatives H_{1n} , it can similarly be shown that the result of Theorem 4 is still formally valid. Then the result of Theorem 6 follows by applying the same argument as in the proof of Theorem 3. We omit the details to avoid repetition.

Appendix B: Lemmas and proofs

Lemma 1 *Given Assumption 3, the $k(\cdot)$ function satisfies*

$$\frac{1}{b} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) \rightarrow \int_0^\infty k(x) dx, \quad \frac{1}{b} \sum_{l=-(n-1)}^{(n-1)} \left| k\left(\frac{l}{b}\right) \right| \rightarrow \int_{-\infty}^\infty |k(x)| dx.$$

Proof First note that for any fixed b ,

$$\begin{aligned} \frac{1}{b} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) &= \frac{n}{b} \sum_{l=1}^{(n-1)} k\left(\frac{l}{n} \times \frac{n}{b}\right) \left(\frac{1}{n}\right) \\ &= \frac{n}{b} \left(\int_0^1 k\left(u \times \frac{n}{b}\right) du + O\left(\frac{1}{n}\right) \right) \end{aligned}$$

where we use the right Riemann sum approximation of the integral over the compact interval $[0, 1]$ (see p.243 of Hughes-Hallett et al., 2009). Now using the change of variable $x = (n/b)u$, we have

$$\frac{1}{b} \sum_{l=1}^{(n-1)} k\left(\frac{l}{b}\right) = \frac{n}{b} \left(\frac{b}{n} \int_0^{n/b} k(x) dx + O\left(\frac{1}{n}\right) \right) = \int_0^{n/b} k(x) dx + O\left(\frac{1}{b}\right) \rightarrow \int_0^\infty k(x) dx$$

by letting $b \rightarrow \infty$ and $n/b \rightarrow \infty$. This completes the proof for the first result. The second result can be proved in a similar way and we omit the details. $\square$

Lemma 2 *Under the same assumptions as Theorem 1,*

$$\bar{d} \xrightarrow{p} \int_0^1 m(x) dx.$$

Proof Note that

$$\bar{d} = \frac{1}{n} \sum_{t=1}^n m_t + \frac{1}{n} \sum_{t=1}^n \varepsilon_t.$$

By the right-point rule of the Riemann integral approximation $\frac{1}{n} \sum_{t=1}^n m_t \rightarrow \int_0^1 m(x) dx$, and also $\frac{1}{n} \sum_{t=1}^n \varepsilon_t \xrightarrow{p} 0$, the result of the lemma follows. $\square$

Lemma 3 *Under the same assumptions as Theorem 1, for any fixed integer $l \geq 0$,*

$$\left| \frac{1}{n} \sum_{t=1}^{n-l} m_t m_{t+l} - \frac{1}{n} \sum_{t=1}^n m_t^2 \right| \leq \mathbb{C} \frac{l}{n}.$$

Proof In this lemma and the following, for ease of exposition we give the proof for the case where there is one discontinuity point d_1 ; the extension to the case of multiple points is straightforward. Note that

$$\left| \frac{1}{n} \sum_{t=1}^{n-l} m_t m_{t+l} - \frac{1}{n} \sum_{t=1}^n m_t^2 \right| \leq \left| \frac{1}{n} \sum_{t=1}^{n-l} m_t m_{t+l} - \frac{1}{n} \sum_{t=1}^{n-l} m_t^2 \right| + \left| \frac{1}{n} \sum_{t=1}^{n-l} m_t^2 - \frac{1}{n} \sum_{t=1}^n m_t^2 \right|.$$

By the boundedness of the $m(\cdot)$ function, it is straightforward to show that the above second term satisfies

$$\left| \frac{1}{n} \sum_{t=1}^{n-l} m_t^2 - \frac{1}{n} \sum_{t=1}^n m_t^2 \right| \leq \mathbb{C} \frac{l}{n}.$$

The first term above can be analysed by a further decomposition as follows:

$$\begin{aligned} & \left| \frac{1}{n} \sum_{t=1}^{n-l} m_t m_{t+l} - \frac{1}{n} \sum_{t=1}^{n-l} m_t^2 \right| \\ & \leq \frac{1}{n} \sum_{t=1}^{n-l} m_t |m_{t+l} - m_t| \\ & = \frac{1}{n} \left(\sum_{t=1}^{[(d_1-l/n)n]} + \sum_{t=[(d_1-l/n)n]+1}^{[(d_1+l/n)n]} + \sum_{t=[(d_1+l/n)n]+1}^{n-l} \right) |m_t| |m_{t+l} - m_t| \\ & \equiv B_1 + B_2 + B_3. \end{aligned}$$

The first sum B_1 is over the continuous region of the $m(\cdot)$ function, and by Lipschitz continuity and the boundedness of $m(\cdot)$ over this region,

$$\frac{1}{n} \sum_{t=1}^{\lfloor (d_1-l/n)n \rfloor} |m_t| |m_{t+l} - m_t| \leq \mathbb{C} \cdot \frac{l}{n} \frac{1}{n} \sum_{t=1}^{\lfloor (d_1-l/n)n \rfloor} 1 = \mathbb{C} \cdot \frac{l}{n}.$$

The second sum B_2 is over the discontinuous region. By the boundedness of $m(\cdot)$,

$$\frac{1}{n} \sum_{t=\lfloor (d_1-l/n)n \rfloor + 1}^{\lfloor (d_1+l/n)n \rfloor} |m_t| |m_{t+l} - m_t| \leq \mathbb{C} \cdot \frac{1}{n} \sum_{t=\lfloor (d_1-l/n)n \rfloor + 1}^{\lfloor (d_1+l/n)n \rfloor} 1 = \mathbb{C} \cdot \frac{l}{n}.$$

The third sum B_3 can be analysed in the same way as the first. In total,

$$\left| \frac{1}{n} \sum_{t=1}^{n-l} m_t m_{t+l} - \frac{1}{n} \sum_{t=1}^{n-l} m_t^2 \right| \leq \mathbb{C} \cdot \frac{l}{n}$$

and the result of this lemma follows. $\square$

Lemma 4 *Under the same assumptions as Theorem 3,*

$$\sum_{t=1}^n (\hat{m}_t - \bar{m}_t)^2 = O_p(h^{-1}).$$

Proof Note that

$$\begin{aligned} \sum_{t=1}^n (\hat{m}_t - \bar{m}_t)^2 &= \sum_{t=1}^n \left(\sum_{s=1}^n w_{t,s} \varepsilon_s \right)^2 \\ &= \sum_{t=1}^n \sum_{s=1}^n w_{t,s}^2 \varepsilon_s^2 + 2 \sum_{t=1}^n \sum_{s=1}^n \sum_{j=1}^{s-1} w_{t,s} w_{t,j} \varepsilon_s \varepsilon_j \\ &\equiv B_1 + B_2. \end{aligned}$$

For B_1 , we have $EB_1 \leq \mathbb{C} \sum_{t=1}^n \sum_{s=1}^n w_{t,s}^2 = \mathbb{C} h^{-1}$ using Assumption 2 and $\max_{t,s} w_{t,s} = O(1/(nh))$. For B_2 ,

$$\begin{aligned} E|B_2| &\leq 2 \sum_{t=1}^n \sum_{s=1}^n \sum_{j=1}^{s-1} w_{t,s} w_{t,j} E|\varepsilon_s \varepsilon_j| \\ &\leq \mathbb{C} \sum_{t=1}^n \sum_{s=1}^n \sum_{j=1}^{s-1} w_{t,s} w_{t,j} |s-j|^{-1-\varepsilon \times (1-2/r)} \\ &= \mathbb{C} \sum_{s=1}^n \sum_{j=1}^{s-1} |s-j|^{-1-\varepsilon \times (1-2/r)} \sum_{t=1}^n w_{t,s} w_{t,j} \\ &= O(h^{-1}) \end{aligned}$$

for $\varepsilon > 0$, where in the second step we use the covariance inequality for an α -mixing sequence (e.g. Proposition 2.8 of Fan and Yao, 2008), and in the last step we use $\max_{1 \leq s, t \leq n} w_{t,s} = O(1/(nh))$, $\sum_{t=1}^n w_{t,s} = O(1)$ and $\sum_{s=1}^n \sum_{j=1}^{s-1} |s-j|^{-1-\varepsilon \times (1-2/r)} = O(n)$ as $n \rightarrow \infty$. In total, $E \sum_{t=1}^n (\hat{m}_t - \bar{m}_t)^2 = O(h^{-1})$ and the claimed result of the lemma follows from Markov's inequality. $\square$

Lemma 5 *Under the same assumptions as Theorem 3,*

$$\begin{aligned} \max_{1 \leq t \leq [(d_1-h)n]} |\bar{m}_t - m_t| &= O(h), \\ \max_{[(d_1-h)n]+1 \leq t \leq [(d_1+h)n]} |\bar{m}_t - m_t| &= O(1), \\ \max_{[(d_1+h)n]+1 \leq t \leq n} |\bar{m}_t - m_t| &= O(h). \end{aligned}$$

Proof There are three results to prove and we start with the first result $\max_{1 \leq t \leq [(d_1-h)n]} |\bar{m}_t - m_t| = O(h)$. Notice that $\bar{m}_t$ is a deterministic term and

$$\bar{m}_t - m_t = \frac{\frac{1}{nh} \sum_{s=1}^n K\left(\frac{s-t}{nh}\right) (m_s - m_t)}{\frac{1}{nh} \sum_{j=1}^n K\left(\frac{s-t}{nh}\right)}.$$

The denominator $\frac{1}{nh} \sum_{s=1}^n K\left(\frac{s-t}{nh}\right)$ is clearly $O(1)$ and nondegenerate, while for the numerator note that

$$\begin{aligned} & \left| \frac{1}{nh} \sum_{s=1}^n K\left(\frac{s-t}{nh}\right) (m_s - m_t) \right| \\ &= \frac{1}{nh} \left| \sum_{s=1}^{[n(d_1-h)]} K\left(\frac{s-t}{nh}\right) (m_s - m_t) \right| + \frac{1}{nh} \left| \sum_{s=[n(d_1-h)]+1}^n K\left(\frac{s-t}{nh}\right) (m_s - m_t) \right| \\ &\equiv A_1 + A_2. \end{aligned}$$

For A_2 , when $s \geq [n(d_1-h)] + 1$, the $m(\cdot)$ function is not continuous but we have $\|m_s - m_t\| \leq \mathbb{C}$ uniformly. We thus have

$$A_2 \leq \frac{C}{nh} \sum_{s=[n(d_1-h)]+1}^n K\left(\frac{s-t}{nh}\right) = \frac{C}{n} \sum_{s=[n(d_1-h)]+1}^n \left(K\left(\frac{s-t}{nh}\right) \frac{1}{h} \right).$$

Since $s \geq [n(d_1-h)] + 1$ and $1 \leq t \leq [(d_1-h)n]$, we have $(s-t)/n \neq 0$. By Assumption 5, $K(u)u^2 \rightarrow 0$ as $u \rightarrow \infty$. It then follows that $K\left(\frac{(s-t)/n}{h}\right) \frac{1}{h} = o(h)$ by noting $1/h \rightarrow \infty$. Also note that there are at most $O(n)$ terms in the summation $\sum_{s=[n(d_1-h)]+1}^n$, and we therefore have $\max_{1 \leq t \leq [(d_1-h)n]} A_2 = o(h)$.

For A_1 , note that since $1 \leq s \leq [(d_1-h)n]$ and $1 \leq t \leq [(d_1-h)n]$, s/n and t/n are both in the continuous interval of the $m(\cdot)$ function. Using the triangle inequality and

the right-point approximation of the Riemann sum, we have

$$\begin{aligned} \max_{1 \leq t \leq [(d_1-h)n]} |A_1| &\leq \max_{1 \leq t \leq [(d_1-h)n]} \frac{1}{nh} \sum_{s=1}^{[n(d_1-h)]} K\left(\frac{s/n - t/n}{h}\right) |m(s/n) - m(t/n)| \\ &= \max_{1 \leq t \leq [(d_1-h)n]} \frac{1}{h} \int_0^{d_1-h} K\left(\frac{u - t/n}{h}\right) |m(u) - m(t/n)| du (1 + o(1)). \end{aligned}$$

Using the change of variable $z = (u - t/n)/h$, the above becomes

$$\begin{aligned} &\max_{1 \leq t \leq [(d_1-h)n]} \int_{\frac{-t/n}{h}}^{\frac{d_1-h-t/n}{h}} K(z) |m(zh + t/n) - m(t/n)| dz (1 + o(1)) \\ &\leq \max_{1 \leq t \leq [(d_1-h)n]} \mathbb{C}h \int_{\frac{-t/n}{h}}^{\frac{d_1-h-t/n}{h}} K(z) |z| dz (1 + o(1)) \end{aligned}$$

which is $O(h)$ by noticing that $\int_{-\infty}^{\infty} K(z) |z| dz < \infty$. The first result is therefore proved.

The second result is a straightforward consequence of the boundedness of the $m(\cdot)$ function, while the third result can be proved in the same way as the first result, and the proof for the lemma is then complete. $\square$

Lemma 6 *Under the same assumptions as Theorem 3,*

$$\frac{1}{n} \sum_{t=1}^n (\bar{m}_t - m_t)^2 = O(h).$$

Proof Decomposing the sum into continuous and discontinuous parts, we obtain

$$\frac{1}{n} \sum_{t=1}^n (\bar{m}_t - m_t)^2 \leq \frac{1}{n} \left(\sum_{t=1}^{[(d_1-h)n]} + \sum_{t=[(d_1-h)n]+1}^{[(d_1+h)n]} + \sum_{t=[(d_1+h)n]+1}^n \right) (\bar{m}_t - m_t)^2.$$

Using Lemma 5 and noting that $(1/n) \sum_{t=1}^{[(d_1-h)n]} 1$ and $(1/n) \sum_{t=[(d_1+h)n]+1}^n 1$ are both $O(1)$, the first and the third terms above are $O(h^2)$. The middle term is $O(h)$ because $(\bar{m}_t - m_t)$ is uniformly $O(1)$ in this region and $(1/n) \sum_{t=[(d_1-h)n]+1}^{[(d_1+h)n]} 1 = O(h)$. In total, the middle term dominates and the claim of the lemma follows. $\square$

Appendix C: Simulating critical values

Finite sample null critical values for DM' (equivalently, DM) can be simulated using the following scheme:

1. Simulate a sample of independent and identically distributed $N(0, 1)$ variables $\{d_t^*\}_{t=1}^n$, where n is the sample size.

2. Calculate the DM' statistic for the simulated sample $\{d_t^*\}_{t=1}^n$, denoting this by $DM_1'^*$. In the calculation of $DM_1'^*$, the same kernels $K(\cdot)$, $k(\cdot)$ and same bandwidths b and h are used as in the calculation of the test statistic.
3. Choose a large positive integer B and repeat steps 1-2 B times to obtain $DM_1'^*$, $DM_2'^*$, ..., $DM_B'^*$.
4. The empirical distribution of the simulated statistics $DM_1'^*$, $DM_2'^*$, ..., $DM_B'^*$ is then the simulated approximation of the null distribution for the sample size n . Critical values can be obtained by calculating the appropriate empirical quantiles of this empirical distribution.

Critical values obtained in this way will deliver correct size for DM' whenever d_t is independent and identically normally distributed with zero mean variation, for given choices of $K(\cdot)$, $k(\cdot)$, b and h . For other d_t specifications (that satisfy Assumptions 1-2), these critical values serve as approximations to the true finite sample critical values. Our simulations show that this simulation scheme can greatly improve the finite sample size of the DM' test. The simulation scheme can also be viewed as a parametric bootstrap procedure to obtain the critical values (see, e.g., Efron and Tibshirani, 1994), as it simulates from a parametric submodel.

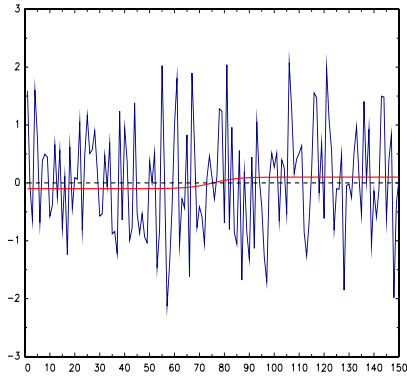
Appendix D: Additional figures

Figure S1 plots generated d_t series, along with the corresponding mean path m_t , for the first Monte Carlo replication of the size simulations discussed in section 5.1 of the paper, for the three cases $a = 0.1, 0.3$ and 0.5 , when $T = 150$ and $\phi = \theta = 0$. Figures S2 and S3 plot d_t and $\hat{m}_t$ for the quarterly UK house price growth forecasts application, for MAE and MSE, respectively. Figures S4 and S5 plot d_t and $\hat{m}_t$ for the quarterly US real output growth forecasts application, for MAE and MSE, respectively.

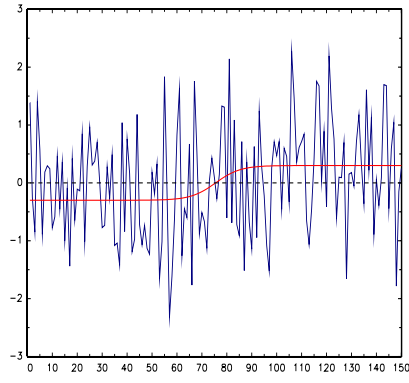
Additional References

- Efron, B. and Tibshirani, R.J. (1994). An Introduction to the Bootstrap. Chapman and Hall, New York.
- Fan, J. and Yao, Q. (2008). Nonlinear Time Series: Nonparametric and Parametric Methods. Springer, New York.
- Hughes-Hallett, D., Lock, P.F., Gleason, A.M. et al. (2009). Applied Calculus. Wiley, New Jersey.

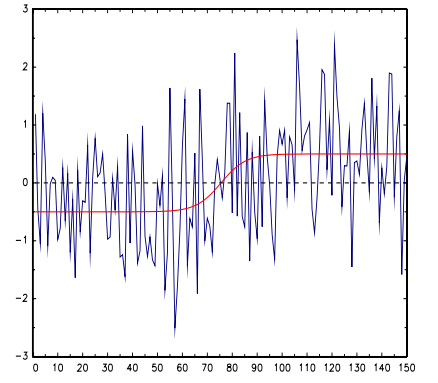
White, H. (2001). Asymptotic Theory for Econometricians (revised edition). Emerald, Bingley.



(a) $a = 0.1$



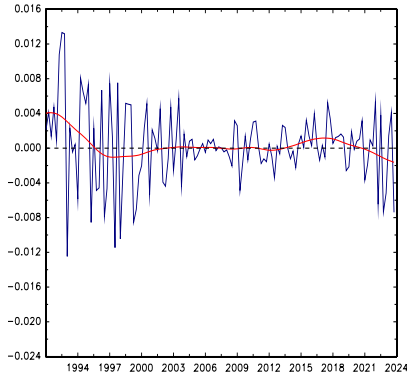
(b) $a = 0.3$



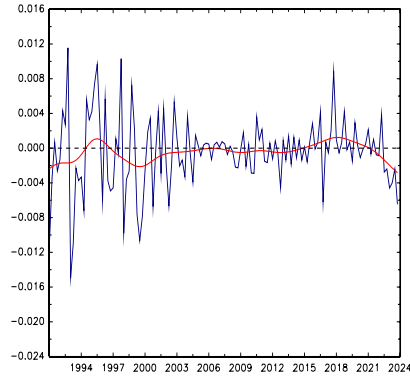
(c) $a = 0.5$

Figure S1. Simulated loss differential and mean function under the null,

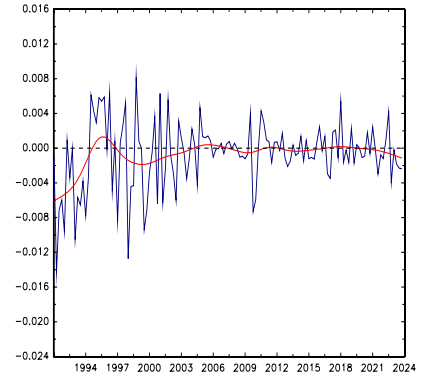
$T = 150, \phi = \theta = 0$: d_t : —; m_t : —



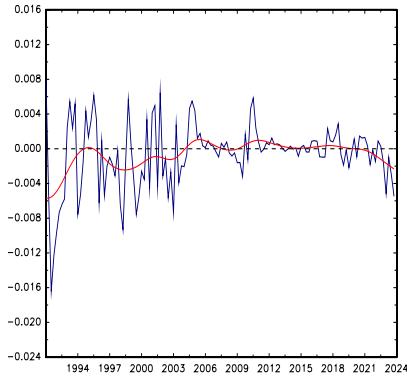
(a) $q = 1$



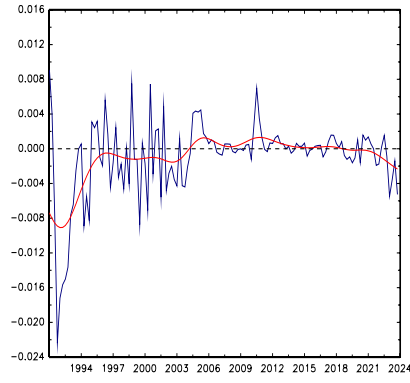
(b) $q = 2$



(c) $q = 3$

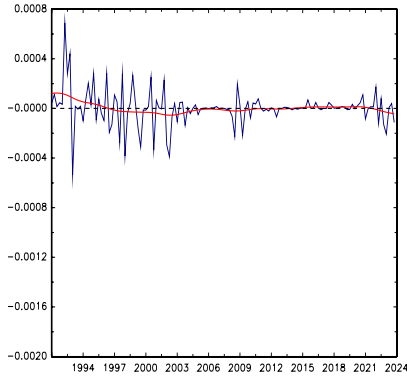


(d) $q = 4$

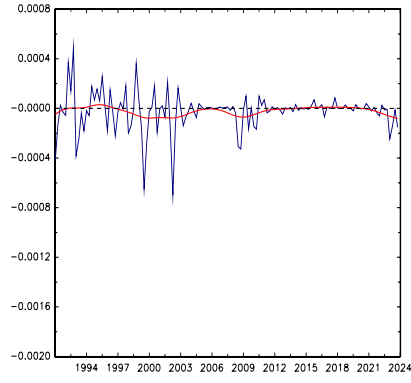


(e) $q = 5$

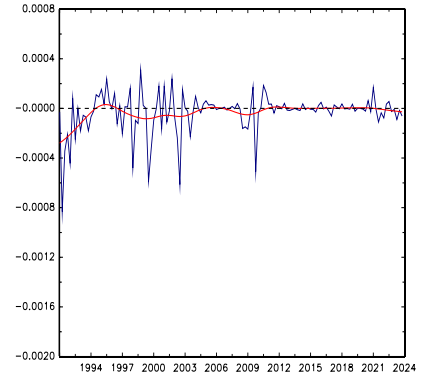
Figure S2. Loss differential and estimated mean function for quarterly UK house price growth forecasts application, MAE: d_t : — ; $\hat{m}_t$: —



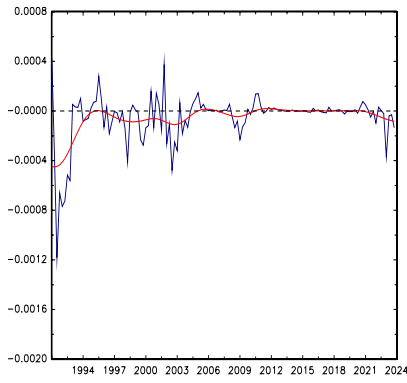
(a) $q = 1$



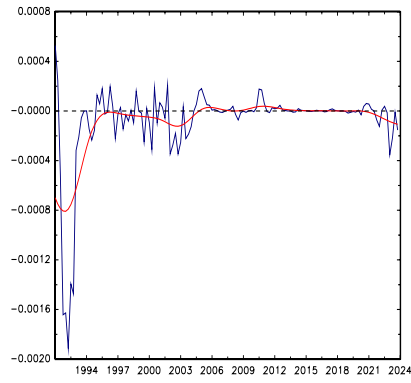
(b) $q = 2$



(c) $q = 3$

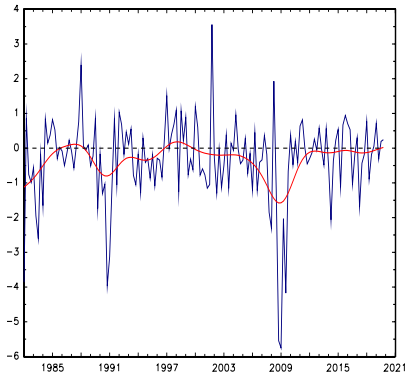


(d) $q = 4$

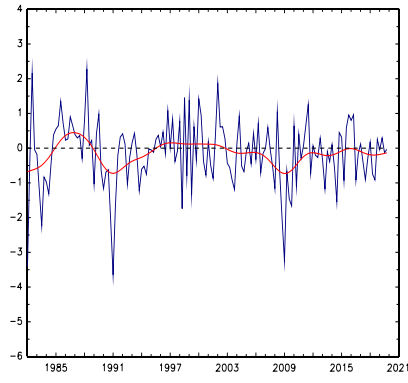


(e) $q = 5$

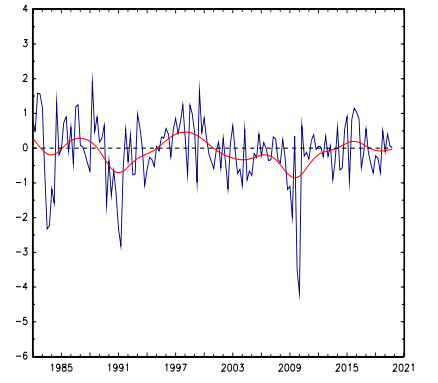
Figure S3. Loss differential and estimated mean function for quarterly UK house price growth forecasts application, MSE: d_t : —; $\hat{m}_t$: —



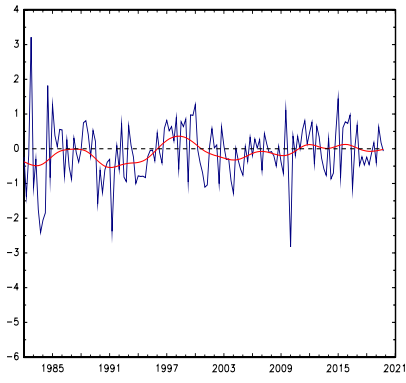
(a) $q = 1$



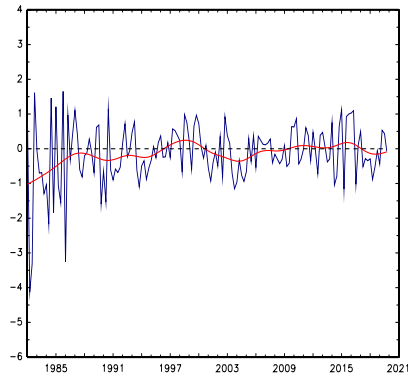
(b) $q = 2$



(c) $q = 3$

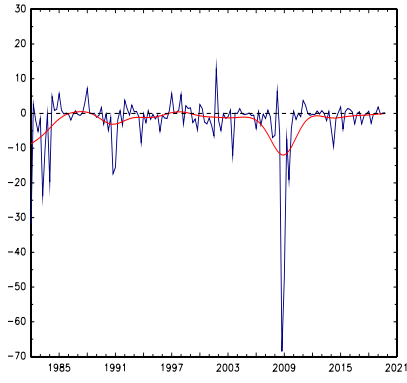


(d) $q = 4$

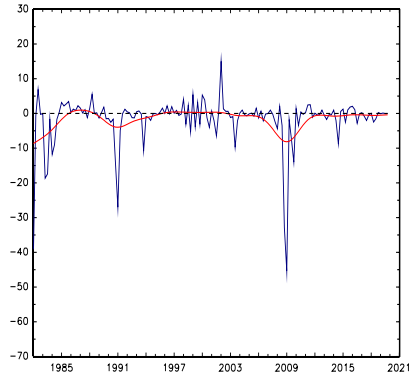


(e) $q = 5$

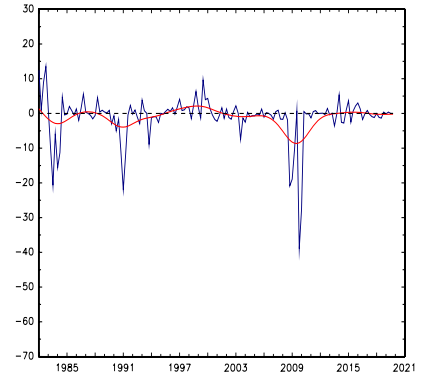
Figure S4. Loss differential and estimated mean function for quarterly US real output growth forecasts application, MAE: d_t : — ; $\hat{m}_t$: —



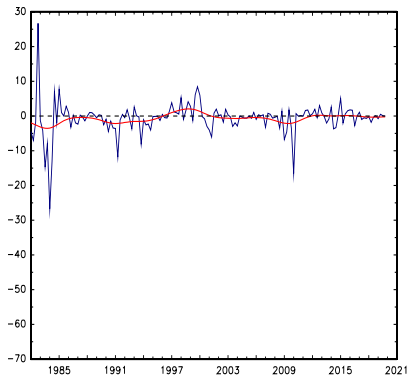
(a) $q = 1$



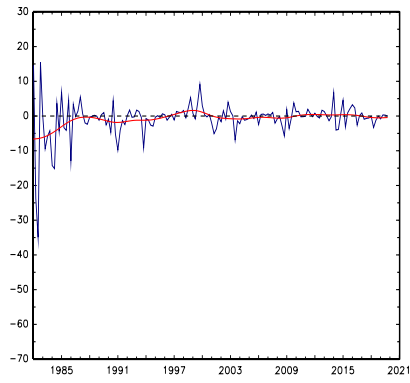
(b) $q = 2$



(c) $q = 3$



(d) $q = 4$



(e) $q = 5$

Figure S5. Loss differential and estimated mean function for quarterly US real output growth forecasts application, MSE: d_t : —; $\hat{m}_t$: —